

Microeconomics **Midterm 2**, Fall 2021

Name and student ID:

Part I: Short Questions (40 points):

1. (5 points) Use Slutsky equation to explain that normal goods must satisfy the law of demand.

$$q_1 = H(P_1, P_2, \bar{U}) = D(P_1, P_2, E(P_1, P_2, \bar{U})) = D(P_1, P_2, Y)$$

$$\frac{\partial H}{\partial P_1} = \frac{\partial D}{\partial P_1} + \frac{\partial D}{\partial E} \times q_1$$

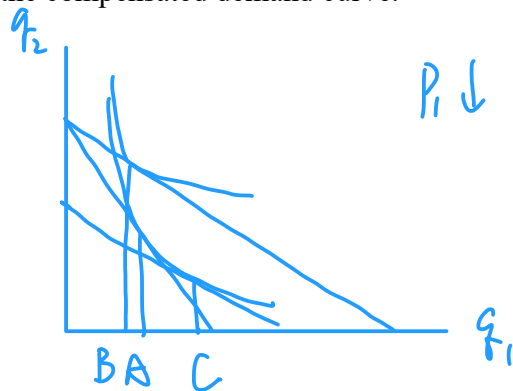
$$\Rightarrow \frac{\partial D}{\partial P_1} = \frac{\partial H}{\partial P_1} - \frac{\partial D}{\partial E} \times q_1 \quad (E=Y)$$

TE SE IE

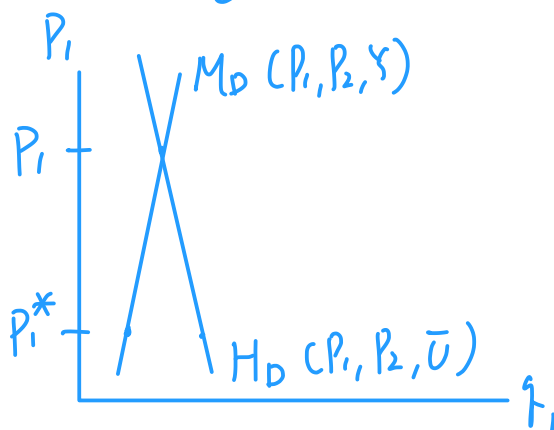
$$\because SE < 0 \quad \cdot \quad \frac{\partial D}{\partial E} > 0 \quad \text{for normal goods}$$

$$\therefore TE = \frac{\partial D}{\partial P_1} < 0 \Rightarrow \text{normal goods satisfy the law of demand}$$

2. (10 points) [True or False] For a Giffen good, the uncompensated demand curve is steeper than the compensated demand curve.



$P_1 \downarrow$ and q_1 is Giffen



True: the slope of $M_D >$
the slope of H_D

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Part I: Short Questions (40 points):

1. (5 points) Use Slutsky equation to explain that normal goods must satisfy the law of demand.

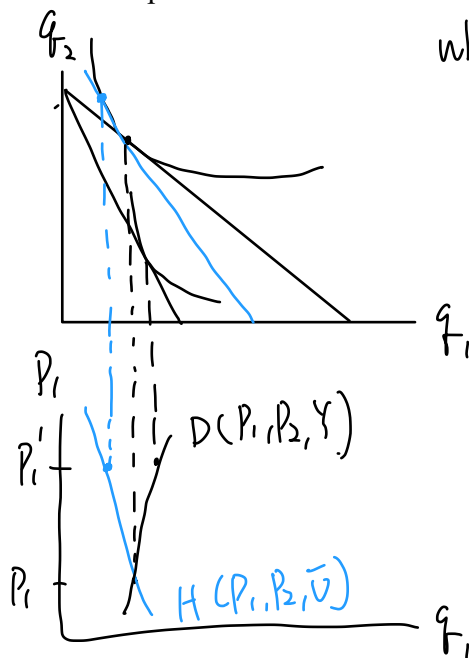
$$\text{Slutsky equation} \Rightarrow TE = SE + IE \quad q_i = H(P_1, P_2, \bar{U}) = D(P_1, P_2, Y) \\ = D(P_1, P_2, E(P_1, P_2, \bar{U}))$$

$$\frac{\partial H}{\partial P_1} = \frac{\partial D}{\partial P_1} + \frac{\partial D}{\partial E} \times \frac{\partial E}{\partial P_1}$$

$$\frac{\partial H}{\partial P_1} = \frac{\partial D}{\partial P_1} + \frac{\partial D}{\partial E} \times q_1 \Rightarrow \frac{\partial D}{\partial P_1} = \frac{\partial H}{\partial P_1} - \frac{\partial D}{\partial E} \times q_1$$

$$\frac{\partial D}{\partial P_1} = TE \quad \frac{\partial H}{\partial P_1} = SE \quad - \frac{\partial D}{\partial E} \times q_1 = IE \quad \because SE < 0, \frac{\partial D}{\partial E} > 0 \text{ for normal goods} \\ \therefore \frac{\partial D}{\partial P_1} < 0 \text{ for normal goods}$$

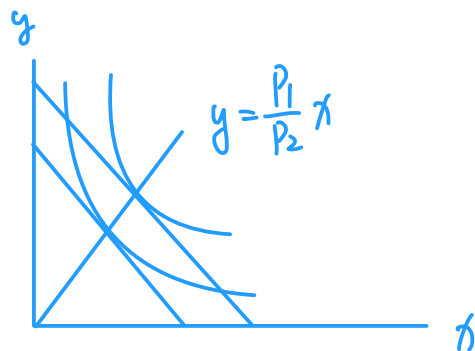
2. (10 points) [True or False] For a Giffen good, the uncompensated demand curve is ~~steeper than~~ the compensated demand curve.



τ : the slope of $D(P_1, P_2, Y) >$ the slope of $H(P_1, P_2, Y)$

larger than the slope of

3. (10 points) Cindy's utility function for goods x and y is $U(x, y) = 0.5 \ln x + 0.5 \ln y$. Plot her income-consumption curve and Engel curve for good x .



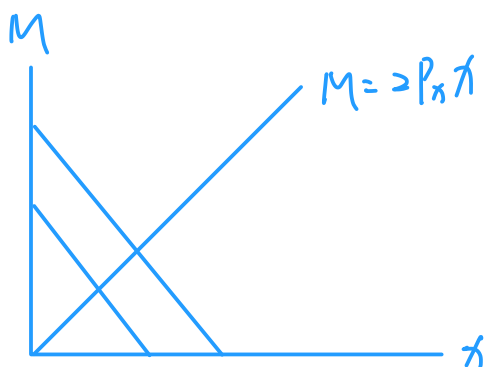
$$MU_x = \frac{1}{2x} \quad MU_y = \frac{1}{2y} \quad MRS = \frac{\frac{1}{2x}}{\frac{1}{2y}} = \frac{y}{x}$$

$$MRS = MRT \quad MRT = \frac{P_x}{P_y} \\ \frac{y}{x} = \frac{P_x}{P_y} \Rightarrow \text{interior sol.}$$

$$P_x x + P_y y = M$$

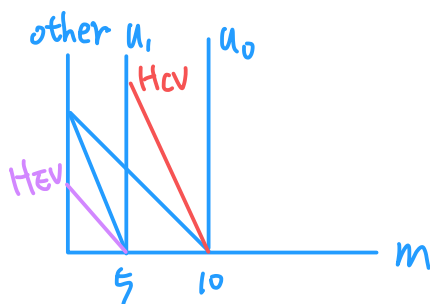
$$P_x x + P_x x = M$$

$$2P_x x = M$$



4. (10 points) Jeremy derives all of his utility from consuming milk shakes; he devotes his entire \$20 allowance to milk shakes each week. Suppose the price of milk shakes rise from \$2 to \$4. Compute Jeremy's compensating variation and equivalent variation.

U = number of milk shakes



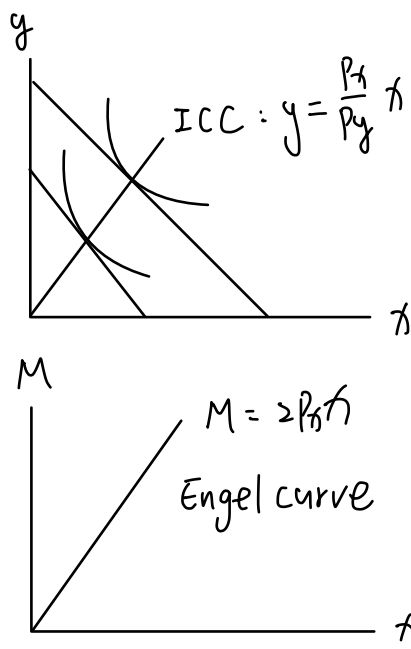
$$u_0 = \frac{20}{2} = 10$$

$$u_1 = \frac{20}{4} = 5$$

$$|CV|: 4 \times 10 - 20 = 20$$

$$|EV|: 2 \times 5 - 20 = 10$$

3. (10 points) Cindy's utility function for goods x and y is $U(x, y) = 0.5 \ln x + 0.5 \ln y$. Plot her income-consumption curve and Engel curve for good x .



$$MU_x = 0.5 \frac{1}{x} = \frac{1}{2x}$$

$$MU_y = 0.5 \frac{1}{y} = \frac{1}{2y}$$

$$MRS = \frac{MU_x}{MU_y} = \frac{\frac{1}{2x}}{\frac{1}{2y}} = \frac{y}{x}$$

$$MRT = \frac{P_x}{P_y}$$

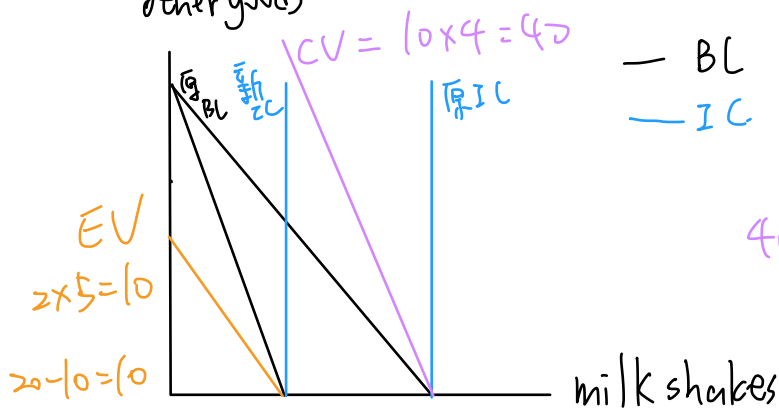
$$MRS = MRT \Rightarrow \frac{y}{x} = \frac{P_x}{P_y} \Rightarrow P_x x = P_y y$$

$$P_x x + P_y y = M$$

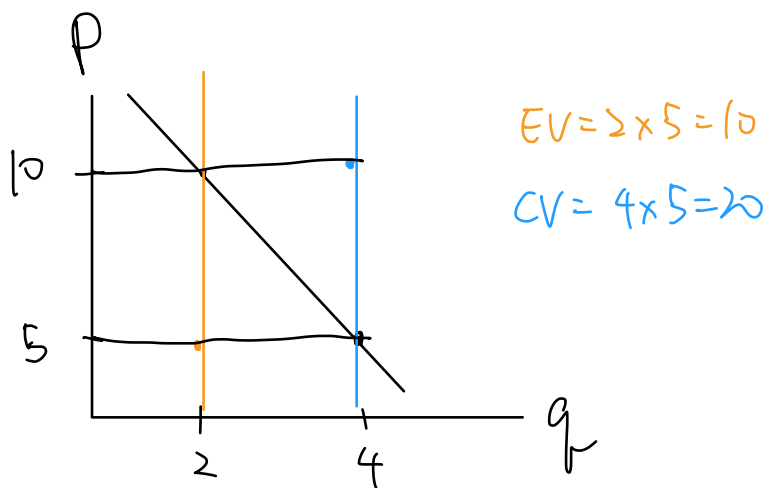
$$2P_y y = M \Rightarrow y = \frac{M}{2P_y} \quad x = \frac{M}{2P_x}$$

4. (10 points) Jeremy derives all of his utility from consuming milk shakes; he devotes his entire \$20 allowance to milk shakes each week. Suppose the price of milk shakes rise from \$2 to \$4. Compute Jeremy's compensating variation and equivalent variation.

other goods



$$\text{utility} = \frac{20}{2} = 10 \text{ milk shakes}$$



5. (5 points) [True or False] The difference between the equivalent variation and compensating variation is greater for goods with larger income elasticities.

slutsky equation: $TE = SE + IE$

$$TE = \frac{\partial D}{\partial P_1} \quad SE = \frac{\partial H}{\partial P_1} \quad IE: -\frac{\partial D}{\partial Y} \times q_1$$

$$\Rightarrow \frac{\partial D}{\partial P_1} \times \frac{P_1}{q_1} = \frac{\partial H}{\partial P_1} \times \frac{P_1}{q_1} - \frac{\partial D}{\partial Y} \times \frac{Y}{q_1} \times \frac{q_1}{Y} P_1$$

elasticity of M_D elasticity of H_D - income elasticity $\times$ fraction of income spent on q_1

由上式知, 当 income elasticity 愈大, 则 CV 与 EV 的差距愈大

$\Rightarrow$ True

Part II: (80 points)

1. (40 points) Alan's preferences over hamburgers (H) and milkshakes (M) are described by the utility function: $U(H, M) = \min(H, 2M)$. His monthly income is Y dollars, and he only buys these two goods out of his income. Denote the price of hamburgers by P_H and of milkshakes by P_M .
- a. (10 points) Derive Alan's uncompensated demand curve for hamburgers and milkshakes as a function of the exogenous variables.

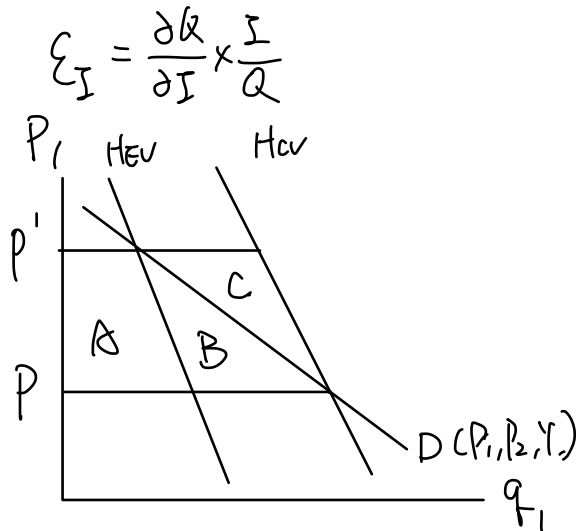
$$U(H, M) = \min(H, 2M) \quad P_H H + P_M M = Y$$

$$\begin{cases} H = 2M \\ P_H H + P_M M = Y \end{cases}$$

$$\Rightarrow P_H 2M + P_M M = Y \Rightarrow M(2P_H + P_M) = Y \Rightarrow M = \frac{Y}{2P_H + P_M}$$

$$\Rightarrow P_H H + P_M \frac{1}{2} H = Y \Rightarrow H(P_H + \frac{1}{2} P_M) = Y \Rightarrow H = \frac{Y}{P_H + \frac{1}{2} P_M}$$

5. (5 points) [True or False] The difference between the equivalent variation and compensating variation is greater for goods with larger income elasticities.



Part II: (80 points)

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 - a. (10 points) Derive Alan's uncompensated demand curve for hamburgers and milkshakes as a function of the exogenous variables.

$$\begin{cases} H = 2M \\ P_H H + P_M M = Y \end{cases}$$

$$P_H \times 2M + P_M M = Y$$

$$M(2P_H + P_M) = Y$$

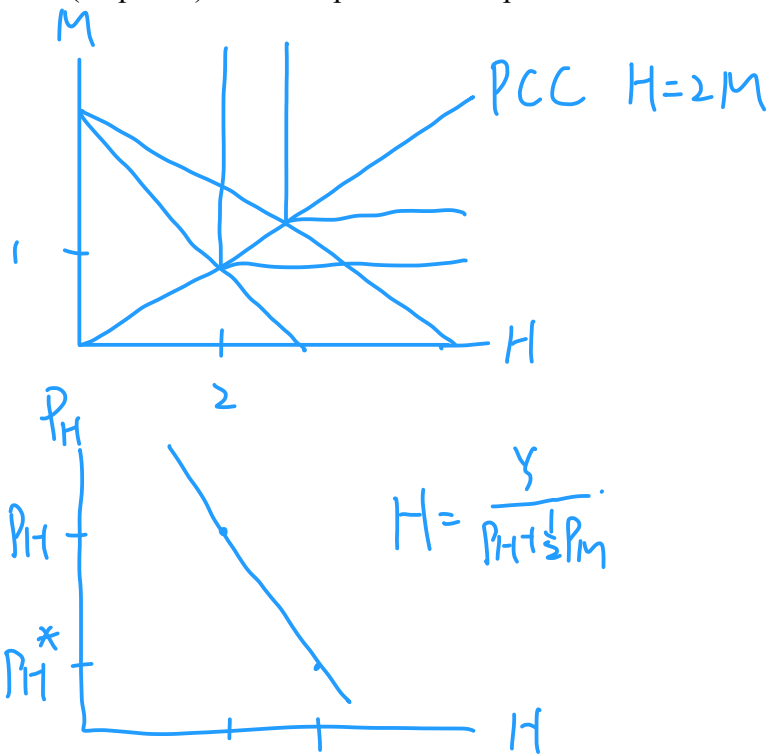
$$M = \frac{Y}{2P_H + P_M} \Rightarrow \text{milkshakes uncompensated demand curve}$$

$$P_H H + P_M \times \frac{1}{2} H = Y$$

$$H(P_H + \frac{1}{2} P_M) = Y$$

$$H = \frac{Y}{P_H + \frac{1}{2} P_M} \Rightarrow \text{hamburgers uncompensated demand curve}$$

- b. (10 points) Draw his price-consumption curve and demand curve for hamburgers.



- c. (10 points) Derive his compensated demand functions for hamburgers and milkshakes and his expenditure function.

$$\min P_H H + P_M M \quad \text{s.t.} \quad \bar{U} = \min(H, 2M)$$

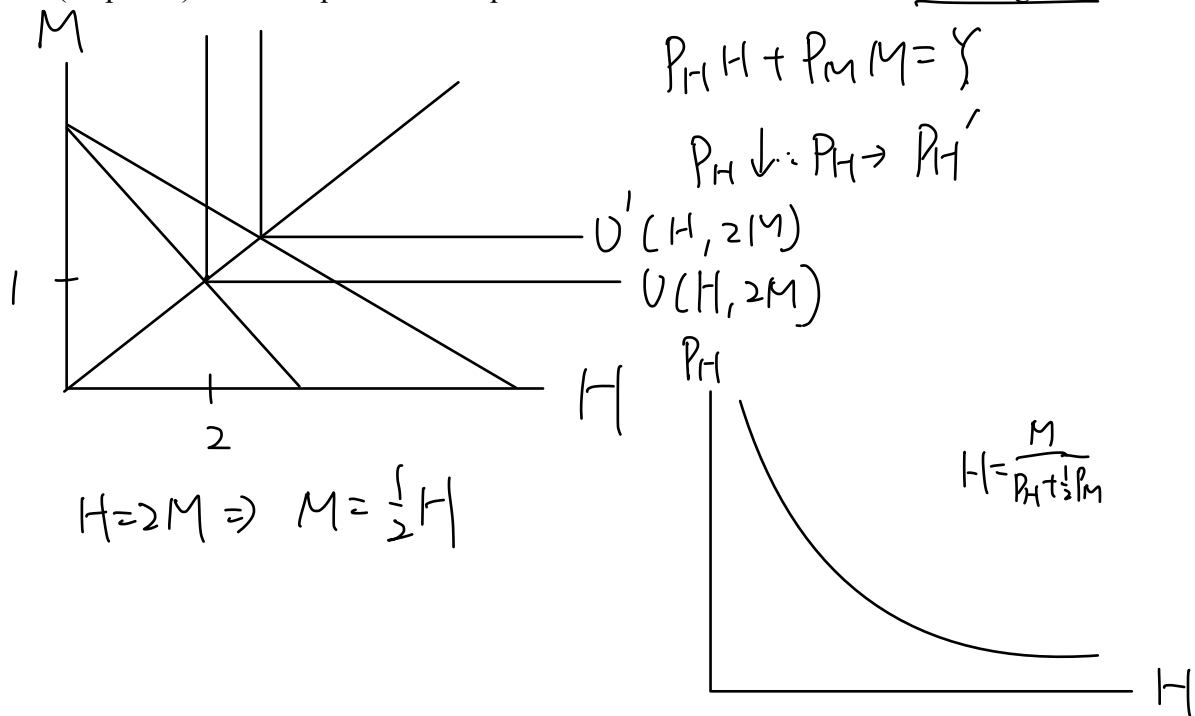
$$\begin{cases} \bar{U} = H \\ \bar{U} = 2M \end{cases}$$

$$E = P_H \times \bar{U} + P_M \times \frac{\bar{U}}{2}$$

$$H_D \text{ for } H : H = \bar{U}$$

$$\text{for } M : M = \frac{1}{2} \bar{U} \quad \#$$

- b. (10 points) Draw his price-consumption curve and demand curve for hamburgers.



- c. (10 points) Derive his compensated demand functions for hamburgers and milkshakes and his expenditure function.

$$\min P_H H + P_M M \quad \text{s.t.} \quad \bar{U} = \min(H, 2M)$$

$$\mathcal{L} = P_H H + P_M M + \lambda(\bar{U} - H)$$

$$P_H H + P_M M + \lambda(\bar{U} - 2M)$$

$$\mathcal{L}_H = P_H - \lambda \Rightarrow \frac{P_H}{P_M} = \frac{1}{2} \quad \geq P_H = P_M$$

$$\mathcal{L}_M = P_M - 2\lambda \Rightarrow$$

$$2\lambda = \bar{U} - H \Rightarrow \bar{U} = H \Rightarrow \text{Hamburger } U_n \dots DC$$

$$\mathcal{L}_\lambda = \bar{U} - 2M = 0 \Rightarrow \bar{U} = 2M \Rightarrow \text{milkshakes } U_n \dots DC$$

$$M = \frac{\bar{U}}{2}$$

$$E = P_H \times \bar{U} + P_M \times \frac{\bar{U}}{2}$$

- d. (10 points) Initially, $Y=120$ and $P_H = P_M = 1$. What are the compensating variation and equivalent variation if P_M increases from 1 to 2?

$$\max U = \min(H, 2M) \quad \text{s.t.} \quad H + M = 120$$

$$\begin{cases} H = 2M \\ H + M = 120 \end{cases} \Rightarrow \begin{cases} 3M = 120 \\ M = 40 \end{cases} \quad H = 80 \quad U = 80$$

$$\min H + 2M \quad \text{s.t.} \quad 80 = \min(H, 2M)$$

$$\begin{cases} 80 = H \\ 80 = 2M \end{cases} \Rightarrow \begin{cases} H = 80 \\ M = 40 \end{cases} \quad E = 80 + 40 \times 2 = 160$$

$$|CV| = |160 - 120| = 40$$

$$\max U = \min(H, 2M) \quad \text{s.t.} \quad H + 2M = 120$$

$$\begin{cases} H = 2M \\ H + 2M = 120 \end{cases} \Rightarrow \begin{cases} 4M = 120 \\ M = 30 \end{cases} \quad H = 60 \quad U = 60$$

$$\min H + M \quad \text{s.t.} \quad 60 = \min(H, 2M)$$

$$\begin{cases} H = 60 \\ 2M = 60 \end{cases} \Rightarrow \begin{cases} H = 60 \\ M = 30 \end{cases} \quad E = 60 + 30 = 90$$

$$|EV| = |90 - 120| = 30 \quad \#$$

- d. (10 points) Initially, $Y=120$ and $P_H = P_M = 1$. What are the compensating variation and equivalent variation if P_M increases from 1 to 2?

$$\max U = (H, 2M) \quad \text{s.t.} \quad H + M = 120$$

$$\begin{cases} H = 2M \\ H + M = 120 \end{cases} \Rightarrow \begin{matrix} H = 80 \\ M = 40 \end{matrix}$$

$$CV: \min H + 2M \quad \text{s.t.} \quad 80 = (H, 2M)$$

$$\begin{cases} 80 = H \\ 80 = 2M \end{cases} \Rightarrow \begin{matrix} H = 80 \\ M = 40 \end{matrix} \quad E = 80 + 2 \times 40 = 160$$

$$|CV| = |160 - 120| = 40$$

$$EV: \max U(H, 2M) \quad \text{s.t.} \quad H + 2M = 120$$

$$\begin{cases} H = 2M \\ H + 2M = 120 \end{cases} \Rightarrow \begin{matrix} M = 30 \\ H = 60 \end{matrix} \quad U = 60$$

$$\min H + M \quad \text{s.t.} \quad 60 = (H, 2M)$$

$$\begin{cases} H = 60 \\ 2M = 60 \end{cases} \Rightarrow \begin{matrix} H = 60 \\ M = 30 \end{matrix} \quad E = 60 + 30 = 90$$

$$EV = |90 - 120| = 30$$

2. (20 points) An economy produces two final goods, X and Y. Their prices and quantities are as follows:

Period	X		Y	
	Quantity	Price	Quantity	Price
1	50	\$4	100	\$3
2	100	\$6	60	\$6

- a. (10 points) Calculate the CPI index in the second period on the assumption that the first period is the base period. What is the weight for each good?

$$CPI_2 = \frac{50 \times 6 + 100 \times 6}{50 \times 4 + 100 \times 3} = 1.8$$

$$\theta_X = \frac{50 \times 4}{50 \times 4 + 100 \times 3} = 0.4$$

$$\theta_Y = \frac{100 \times 3}{50 \times 4 + 100 \times 3} = 0.6$$

- b. (10 points) In the textbook, the author states that the CPI index tends to overestimate the true cost of price increases. Why? Does this statement hold for all kinds of consumer preferences and all kinds of price increases? Explain.

∴ CPI adj. 沒有考慮 substitution effect, 當一商品 P ↑, consumer 會用便宜的商品替代貴的商品

在商品 P ↑ 幅度相同時, 因為沒有替代效應
或完全互補品時也沒有替代效果, 因為兩者一定以
固定比例購買

2. (20 points) An economy produces two final goods, X and Y. Their prices and quantities are as follows:

	X		Y	
Period	Quantity	Price	Quantity	Price
1	50	\$4	100	\$3
2	100	\$6	60	\$6

- a. (10 points) Calculate the CPI index in the second period on the assumption that the first period is the base period. What is the weight for each good?

$$CPI = \frac{50 \times 6 + 100 \times 6}{50 \times 4 + 100 \times 3} = \frac{900}{500} = 1.8$$

$$X: \frac{50 \times 4}{500} = 0.4$$

$$Y_1 = 50 \times 4 + 100 \times 3$$

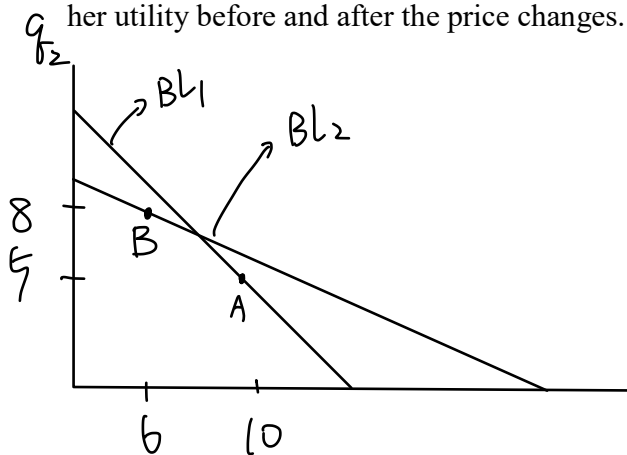
$$Y_2 = 50 \times 6 + 100 \times 6$$

$$Y = \frac{100 \times 3}{500} = 0.6$$

- b. (10 points) In the textbook, the author states that the CPI index tends to overestimate the true cost of price increases. Why? Does this statement hold for all kinds of consumer preferences and all kinds of price increases? Explain.

3. (20 points) Answer the following two questions.

- a. (10 points) Remy spends her weekly income of \$30 on chocolate, q_1 , and shampoo, q_2 . Initially, when the prices are $p_1=p_2=2$, she buys $q_1=10$ and $q_2=5$. After the prices change to $p_1=1$ and $p_2=3$. She purchases $q_1=6$ and $q_2=8$. Draw her budget lines and choices in a diagram. Use a revealed preference argument to discuss whether or not she is maximizing her utility before and after the price changes.

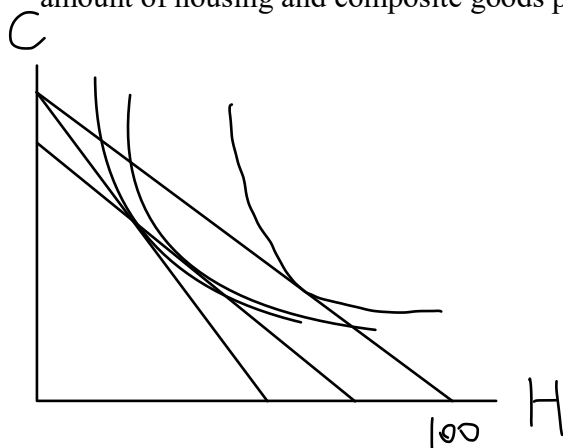


沒有效用極大化
因為在價格改變前 B 也是可達到的
但是沒有選，而 P 改變後 A 也是可以
而這次選 B，則若 Remy 對 A RP B
在 P 改變後也應該選 A，若 Remy
對 B RP A 在 P 改變前也應該選 B

$\Rightarrow$ Remy is not maximizing utility

	A: $q_1=10$ $q_2=5$	B: $q_1=6$ $q_2=8$
A: $p_1=2$ $p_2=2$	30	$\geq 8^*$
B: $p_1=1$ $p_2=3$	25^*	30

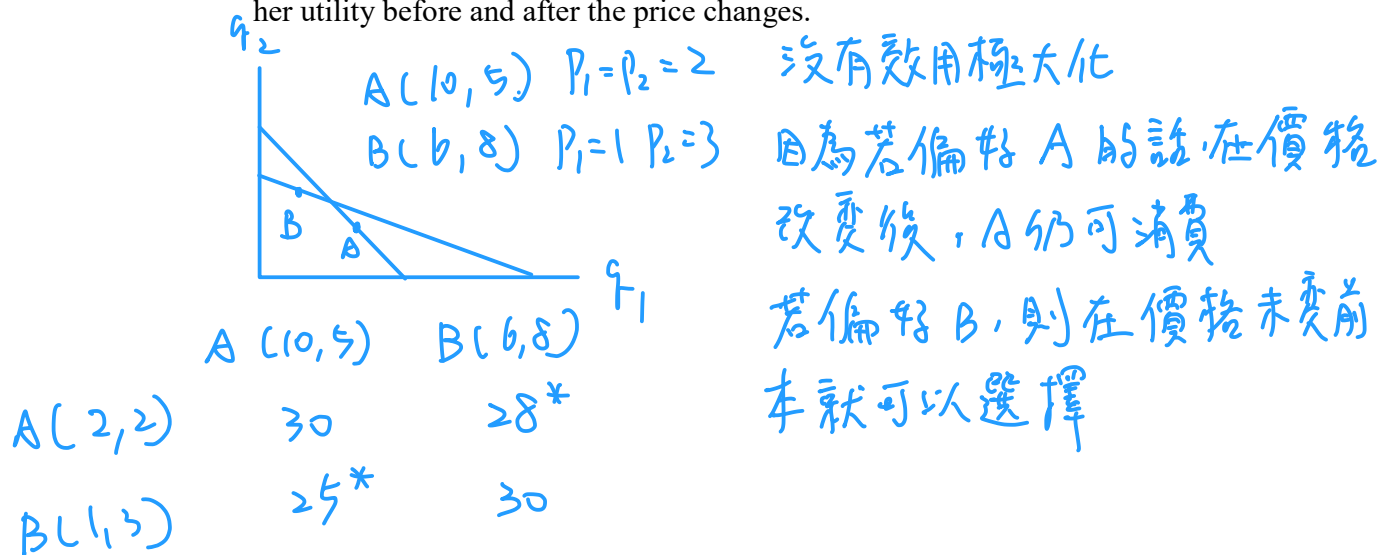
(10 points) A consumer purchases housing (H) and spends the remainder of income on the composite good (C). The government is considering one of two policies. Policy A taxes housing by \$50 per unit consumed. With the tax in place, the consumer purchases 100 units of housing. Policy B collects a lump-sum tax of \$5,000 from the consumer's income. Compare the effects of the policies on the consumer's utility/well-being and the amount of housing and composite goods purchased.



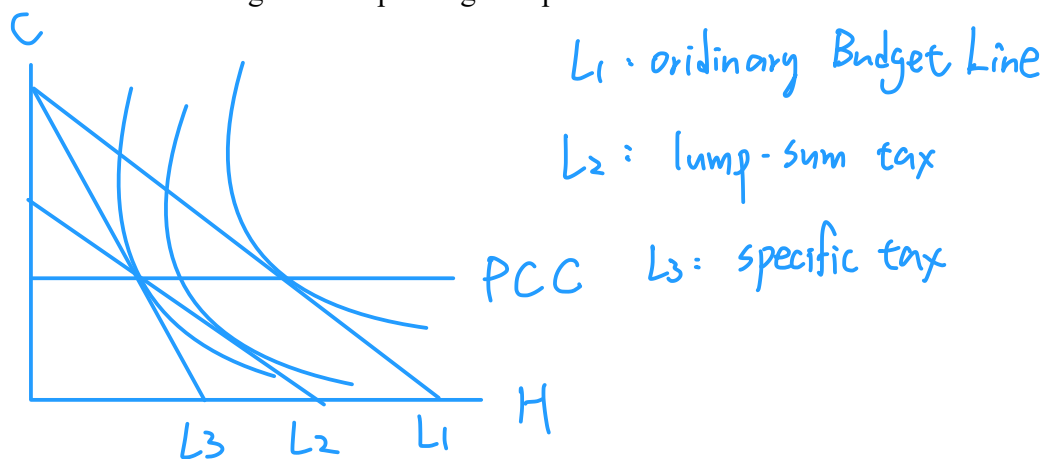
$$P_H H + P_C C = I$$

3. (20 points) Answer the following two questions.

- a. (10 points) Remy spends her weekly income of \$30 on chocolate, q_1 , and shampoo, q_2 . Initially, when the prices are $p_1=p_2=2$, she buys $q_1=10$ and $q_2=5$. After the prices change to $p_1=1$ and $p_2=3$. She purchases $q_1=6$ and $q_2=8$. Draw her budget lines and choices in a diagram. Use a revealed preference argument to discuss whether or not she is maximizing her utility before and after the price changes.



- b. (10 points) A consumer purchases housing (H) and spends the remainder of income on the composite good (C). The government is considering one of two policies. Policy A taxes housing by \$50 per unit consumed. With the tax in place, the consumer purchases 100 units of housing. Policy B collects a lump-sum tax of \$5,000 from the consumer's income. Compare the effects of the policies on the consumer's utility/well-being and the amount of housing and composite goods purchased.



以 Cobb-Douglas 函數作假設 $U = H^a C^{1-a}$

Policy A 為 specific tax 可減少較多 consumer 對 H 的消費

Policy B 為 lump-sum tax 減少較少的 H 消費量, 但 consumer 的 utility 也較高